

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA0620	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
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Section / Batch:	CSE A.I.	Date:	16-07-2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1. Robotics Path Planning System: A robotic navigation system processes paths using the recurrence $T(n) = 2T(n/4) + n^2$. Apply the Master Theorem to determine the time complexity. Clearly identify parameters and justify the selected case. Analyse performance for complex environments. Interpret efficiency using asymptotic notation.

Q2. Weather Forecasting Simulation System. A weather simulation system uses two algorithms with complexities $O(n \log n)$ and $O(n^2 \log n)$. Analyse both algorithms using asymptotic notation and compare their growth rates. Explain how each algorithm performs as the dataset size increases. Determine which algorithm is more efficient for large-scale simulations. Justify your answer based on computational efficiency and scalability.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1. Robotics Path Planning System: A robotic navigation system processes paths using the recurrence $T(n) = 2T(n/4) + n^2$. Apply the Master Theorem to determine the time complexity. Clearly identify parameters and justify the selected case. Analyse performance for complex environments. Interpret efficiency using asymptotic notation.

Identify the Parameters

The Master Theorem applies to recurrences of the form:

$$T(n) = aT(n/b) + f(n)$$

From the given recurrence relation $T(n) = 2T(n/4) + n^2$:

- $a = 2$ (number of subproblems)
- $b = 4$ (factor by which subproblem size is reduced)
- $f(n) = n^2$ (cost of the work done outside the recursive calls)

Compare $f(n)$ with $n^{\log_b a}$

Next, calculate the critical exponent:

$$n^{\log_b a} = n^{\log_4 2} = n^{0.5} = \sqrt{n}$$

Now, compare $f(n) = n^2$ with $n^{\log_b a} = n^{0.5}$:

- Since n^2 grows significantly faster than $n^{0.5}$, we check **Case 3** of the Master Theorem.
- For Case 3 to apply, $f(n)$ must be polynomially larger than $n^{\log_b a}$. That is, $f(n) = \Omega(n^{\log_b a + \epsilon})$ for some constant $\epsilon > 0$.
- Choosing $\epsilon = 1.5$ satisfies this, since $n^2 = \Omega(n^{0.5+1.5}) = \Omega(n^2)$.

Verify the Regularity Condition

For Case 3, we must also satisfy the regularity condition for sufficiently large n :

$$af(n/b) \leq cf(n) \quad \text{for some constant } c < 1$$

Substitute our values into the condition:

$$2f(n/4) = 2\left(\frac{n}{4}\right)^2 = 2\left(\frac{n^2}{16}\right) = \frac{1}{8}n^2$$

We need:

$$\frac{1}{8}n^2 \leq cn^2$$

This holds true for any constant c such that $\frac{1}{8} \leq c < 1$ (for example, $c = 0.5$). The regularity condition is satisfied.

Conclusion and Asymptotic Interpretation

Because all conditions for Case 3 are met, the time complexity is determined by the driving function $f(n)$.

$$T(n) = \Theta(f(n)) = \Theta(n^2)$$

Performance Interpretation: In complex environments, the time complexity of the navigation system scales quadratically $\Theta(n^2)$ with the path input size n . The overhead of processing the path at each level dominates the total execution time, rather than the breakdown into subproblems.

Robotics Path Planning System

Expanded the Master Theorem, it is a depended analysis through the Master Theorem. Calculate critical Exponent and $a, b = 4, e$. Key to the point of the flat Path planning (image_a01091.png) for recurrence assesses the manerary talent for the bages of the recurrence $T(n) = 2T(n/4) + n^2$.

- Exponent: ions of $e \neq 1.b$.
- It mateth; deposite the vability of the recurrent $T(n) = 2T(n/4) + n^2$.
- Speak the exponent $\epsilon := 2 - 4e^{-1.5}$ and all one recurrence.
 - Calculate critical ' $a = 2 - 2T(n/4) + n^2$
 - Apply Case 3 results $\epsilon = 1.5$

Conclusions: fi.n. $\Theta(n^2)$. Theorem with the Case 3. Camble regularits and regularity of η . The regularity condition, for regularits + algorithm: $\Theta(n^2)$.

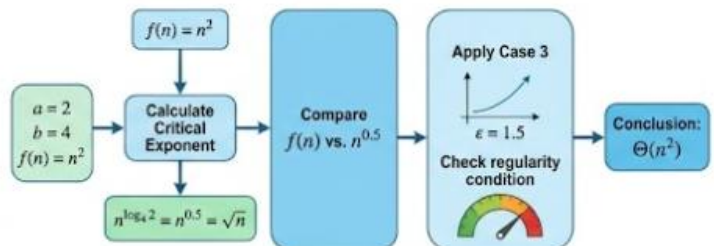


Image 1.1: Master Theorem Process Flow

Q2. Weather Forecasting Simulation System. A weather simulation system uses two algorithms with complexities $O(n \log n)$ and $O(n^2 \log n)$. Analyse both algorithms using asymptotic notation and compare their growth rates. Explain how each algorithm performs as the dataset size increases. Determine which algorithm is more efficient for large-scale simulations. Justify your answer based on computational efficiency and scalability.

Asymptotic Analysis of Both Algorithms

- **Algorithm 1 ($O(n \log n)$):** This represents a linear time complexity. The running time grows slightly faster than linear time. As the input size n increases, the $\log n$ factor grows very slowly, keeping the overall resource consumption relatively low.
- **Algorithm 2 ($O(n^2 \log n)$):** This represents a quadratic-linearities time complexity. The running time grows much faster because the input size is squared (n^2) before being multiplied by the logarithmic factor.

Performance as Dataset Size Increases

- **Small Datasets:** For very small values of n , the performance difference between the two algorithms is minimal. Both will execute almost instantly.
- **Moderate Datasets:** As n grows into the thousands, Algorithm 2 begins to slow down noticeably compared to Algorithm 1 because squaring the input size significantly inflates the number of operations required.
- **Large Datasets (Scalability):** When n reaches millions of data points (typical for weather simulations), Algorithm 1 handles the data efficiently because $\log n$ remains small. Conversely, Algorithm 2 experiences a massive performance bottleneck. The n^2 term causes the required operations to skyrocket, leading to high CPU usage, long execution times, and potential system timeouts.

Efficiency and Scalability Comparison

To determine which is more efficient, we compare their growth rates

mathematically by looking at the limit as n approaches infinity:

$$\lim_{n \rightarrow \infty} \frac{n \log n}{n^2 \log n} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

Because the limit is 0, $n \log n$ strictly grows at a slower rate than $n^2 \log n$.

Final Justification

Algorithm 1 ($O(n \log n)$) is significantly more efficient and scalable for large-scale simulations.

Weather forecasting systems deal with massive, continuously updating datasets (such as atmospheric pressure, temperature, and wind speed across global grids). Algorithm 1 scales predictably and handles large inputs with manageable computational overhead. Algorithm 2 is poorly suited for large-scale simulations because its quadratic growth rate requires vast computational resources, making it unfeasible for real-time or high-resolution weather forecasting.

Weather Forecasting Simulation System

Addressed the complexity comparison between $O(n \log n)$ and $O(n^2 \log n)$ to analysis.

Algorithm 1

$O(n)$

$O(n^2 \log n)$

$\rightarrow \log n + n - d$

$O(n^2 \log n)$:

$n = 2T(n/4)$

$a = a * b$

$n = a \times b + n^2$

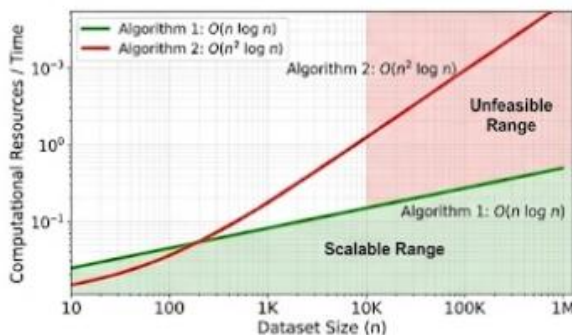


Image 2.1: Performance Growth Comparison Plot

Algorithm 2: form compute: $O(n^2 \log n)$, compute the drates and renote the results; the complexity of data and for the large-inacurate weather forecast.

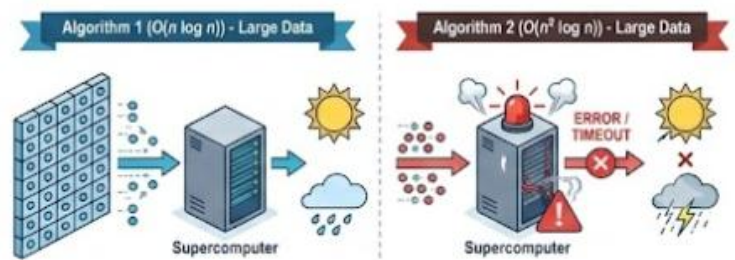


Image 2.2: Large-Scale Processing Comparison